

Denoiser–Representation Interaction in Noise-Robust ESC: A Systematic Study of Classical and Learned Enhancement Methods

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Abstract— Environmental sound classification (ESC) systems often experience significant performance degradation under real-world noise conditions. While numerous denoising approaches have been proposed, the interaction between audio enhancement strategies and downstream representations remains insufficiently explored. This work investigates representation-aware denoising for noise-robust ESC by evaluating three architecturally distinct classification pipelines using the ESC-50 dataset: a raw waveform-based 1D Convolutional Neural Network (CNN), an MFCC-based Subspace Discriminant Ensemble, and an Audio Set-pretrained CNN14 model using log-mel spectrogram representations. Models are evaluated under additive white Gaussian noise (AWGN) and real-world traffic noise across SNR levels ranging from -10 to $+10$ dB. Classical and learning-based enhancement strategies are investigated, including adaptive Wiener filtering, MMSE-LSA, and U-Net-based denoising. Experimental results demonstrate that enhancement performance depends strongly on the interaction between the denoising method and the selected audio representation. The MFCC feature representation benefits from adaptive Wiener filtering, while MMSE-LSA reduces its performance. This suggests that applying spectral enhancement without considering feature extraction may not always be effective. Under stationary noise conditions, MMSE-LSA improves CNN14 performance at low SNR levels, demonstrating its effectiveness when noise stationarity assumptions are held. Notably, MMSE-LSA fails under non-stationary traffic noise due to violated stationarity assumptions, whereas the U-Net denoiser generalizes robustly across both noise types and all three pipelines. The strongest pipeline CNN14 achieves $86.95 \pm 2.5\%$ clean accuracy and recovers up to 24.30 % points at -5 dB SNR for traffic noise using U-Net denoiser. These results demonstrate that effective noise-robust ESC requires joint consideration of the denoising strategy, noise characteristics, and downstream representation rather than treating it as an independent preprocessing step.

Index Terms—Environmental sound classification, noise robustness, U-Net, MMSE-LSA, Wiener filtering, log-mel spectrogram, denoiser–representation interaction.

I. INTRODUCTION

Environmental sound classification (ESC) plays a critical role in a wide range of applications, including smart environments, autonomous systems, urban monitoring, and human-computer interaction. Unlike speech recognition, ESC remains inherently challenging due to the highly variable and non-stationary nature of real-world acoustic events. The ESC-50 dataset [1] has emerged as a widely adopted benchmark for evaluating ESC systems under controlled experimental settings.

A major challenge in practical ESC deployment is the presence of background noise, which degrades both signal quality and classification performance. Although modern deep learning architectures such as CNN14 [2] achieve strong accuracy on clean audio, their robustness under noisy conditions remains limited. Numerous enhancement techniques have been proposed, ranging from classical DSP approaches to learning-based denoisers. However, these methods are typically evaluated independently of the downstream audio representation, overlooking the fact that enhancement artifacts can distort the very acoustic features a classifier depends on.

In this work, we investigate feature representation-based denoising for noise-robust ESC across three architecturally distinct pipelines based on raw waveforms, Mel-Frequency Cepstral Coefficients

(MFCCs), and log-mel spectrograms. Pipelines are evaluated under additive white Gaussian noise (AWGN) and real-world traffic noise across a range of SNR levels. Both classical DSP-based enhancement methods and a learned U-Net denoiser [3]–[4] are examined.

The main contributions of this paper are summarized as follows:

- We demonstrate that denoising effectiveness in ESC is dependent on the underlying audio representation.
- We evaluated classical and learning-based denoising methods across three architecturally distinct ESC pipelines under both synthetic and real-world noise conditions.
- We show that denoising strategies do not generalize uniformly across representations, highlighting the necessity of jointly considering enhancement and feature representation in noise-robust ESC system design.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the experimental setup. Section IV presents the proposed methodology. Section V presents the experimental results, Section VI discusses the findings, and Section VII concludes the paper.

II. RELATED WORK

A. Deep Learning for Environmental Sound Classification

Piczak [1] established the ESC-50 benchmark, demonstrating that Convolutional Neural Networks (CNNs) operating on log-mel spectrograms outperform traditional MFCC-based approaches for environmental sound classification (ESC). Building on this, Kong et al. [2] introduced Pretrained Audio Neural Networks (PANNs), where CNN14 architecture achieved state-of-the-art performance via large-scale AudioSet pretraining, establishing transfer learning as a dominant paradigm in ESC. Subsequent research has further enhanced performance by integrating multi-feature fusion [5] and attention mechanisms [6]. However, this validation relied on clean audio datasets or synthetic environments with minimal noise. Recently, investigation has shifted toward transformer models, self-supervised audio features, and domain generalization techniques to enhance robustness against acoustic covariate shift [7]–[10].

B. Classical Speech Enhancement and Audio Preprocessing

Classical audio enhancement techniques including spectral subtraction, Wiener filtering, and MMSE-based estimators were originally engineered to improve perceptual speech quality rather than optimize downstream classification accuracy. Among these, the decision-directed MMSE-LSA estimator proposed by Ephraim and Malah [11] remains widely adopted for single-channel noise suppression due to its capacity to mitigate musical noise artifacts while preserving structural signal components.

Critically, several studies note that enhancement-induced distortions can degrade downstream classification performance even when perceptual quality improves [12]. This discrepancy indicates that denoising performance varies significantly across different feature representations and model architectures, a relationship that has yet to be systematically characterized.

To address the limitations of statistical estimators, recent literature has explored deep learning-based denoisers [13]–[14], such as U-Net encoder-decoder architectures [3]–[4]. Unlike MMSE-LSA, which assumes locally stationary Gaussian noise, learned denoisers adapt to diverse, non-stationary noise conditions through data-driven training. Nevertheless, classical digital signal processing (DSP) methods remain highly relevant due to their inherent interpretability and low computational overhead.

C. Research Gap

While prior work has advanced ESC architectures and audio enhancement independently, the joint interaction between denoising strategies, feature representations, and noise stationarity remains poorly understood [15]. Classical methods are traditionally evaluated using perceptual metrics rather than classification-oriented performance indicators [16]. Furthermore, while isolated studies suggest that denoising effectiveness depends on the downstream architecture, systematic comparisons

spanning fundamentally distinct representation paradigms namely raw waveforms, cepstral features, and time-frequency spectrograms under both stationary and non-stationary noise are conspicuously limited [15], [17].

This work addresses these gaps by jointly evaluating classical DSP-based and learned denoising strategies across three architecturally distinct ESC pipelines. Our findings demonstrate that achieving noise-robust ESC requires a unified approach that is both representation-aware and noise-type-aware.

III. EXPERIMENTAL SETUP

A. ESC-50: dataset for environmental sound classification

The ESC-50 benchmark [1] is a labeled collection of 2,000 environmental audio clips organized into 50 semantic classes, with 40 samples per class. Each recording is 5 seconds long, sampled at 44.1 kHz in mono. The dataset provides five predefined cross-validation folds to ensure reproducible evaluation, with clips from the same source file restricted to a single fold. The dataset is class-balanced, eliminating the need for class weighting and yielding a chance-level accuracy of 2%. Due to the limited number of samples per class, overfitting remains a significant challenge, motivating the use of data augmentation strategies. The dataset includes diverse sound types ranging from impulsive events (e.g., dog bark, fireworks) to continuous acoustic events (e.g., rain, wind), requiring models to capture both temporal and spectral characteristics effectively. *Table I*, shows the dataset summary.

TABLE I: ESC-50 Dataset Summary

Property	Value
Total Clips	2,000
Number of Classes	50
Samples per Class	40
Clip Duration	5 seconds
Sampling Rate	44.1 kHz (mono)
Cross-validation Folds	5 (400 clips each)
Class Balance	Perfectly balanced
Chance Accuracy	2%

B. Noise Corruption and Evaluation Protocol

All clips were loaded in mono and RMS-normalized to a target level of 0.1 prior to noise addition. Two noise conditions were considered: additive white Gaussian noise (AWGN) and real-world non-stationary traffic noise. AWGN was generated as a zero-mean Gaussian random process with a flat power spectral density, providing a controlled synthetic noise condition for evaluating model behavior under varying signal degradations. The traffic noise used in this study was sourced from Freesound.org [18], an open-access repository of real-world acoustic recordings. Specifically, the recording titled '*Crowd traffic under bridge in Cairo*' was used, captured as an urban field recording in Cairo, Egypt. The original file was a mono M4A recording sampled at 48,000 Hz with a duration of approximately 17.7 seconds. Prior to use, the file was converted to WAV format to ensure compatibility with the pipeline's audio processing framework, resampled to match the ESC-50 sample rate, and segmented into 5-second clips corresponding to the dataset's fixed clip duration. These

clips were then mixed with clean ESC-50 samples at varying SNR levels to simulate realistic non-stationary background interference. The noise signals were scaled and mixed with the clean environmental sound recordings to achieve target signal-to-noise ratio (SNR) levels of $\{-10, -5, 0, 5, 10\}$ dB, computed as in (1):

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \quad (1)$$

Where P_{signal} and P_{noise} denote the power of the clean audio signal and the added noise component, respectively. Corrupted waveforms were clipped to $[-1, 1]$ to prevent overflow. This protocol was applied uniformly across all three pipelines, ensuring a fair inter-pipeline comparison.

Statistical significance of denoising-induced accuracy gains was assessed using paired t-tests (and Wilcoxon signed-rank tests where applicable) across five independent runs per SNR level, with significance level at $p < 0.05$.

IV. METHODOLOGY

A. Pipeline 1: Raw Waveform-Based 1D CNN

The proposed raw waveform based 1D CNN architecture is illustrated in Fig. 1. Audio recordings, originally sampled at 44.1 kHz, were downsampled to 22.05 kHz and segmented into 5-second clips, resulting in input sequences of 110,250 samples. The raw waveform was provided directly to the network as a $110,250 \times 1$ input tensor, allowing the model to learn discriminative representations without relying on hand-crafted features.

The architecture comprises four convolutional blocks containing $\{32, 64, 128 \text{ and } 128\}$ filters with kernel sizes of $\{160, 16, 8 \text{ and } 4\}$, respectively applying a stride of 4 in the initial block. Each block integrates Batch Normalization, ReLU activation, Max Pooling, Squeeze-and-Excitation channel-attention mechanism, and Dropout ($p = 0.30$). A Global Average Pooling layer subsequently feeds into a Dense (128)–ReLU–Dropout (0.5)–Dense (50) output layer with softmax activation.

Training augmentation incorporated circular time-shifting ($\pm 10\%$), amplitude scaling within the range $[0.7, 1.3]$ paired with re-RMS normalization, time masking (two independent segments, each up to 15% of the clip length), and waveform Mixup ($\lambda \sim \text{Beta}(0.4, 0.4)$, $p = 0.5$). Label smoothing ($\epsilon = 0.1$) was applied. Optimization was performed using Adam ($\text{lr} = 10^{-3}$), batch size 32, and early stopping (patience = 7).

For noise robustness evaluation, the raw waveform pipeline was evaluated under two alternative enhancement strategies for comparison. The first strategy is the decision-directed MMSE-LSA (Minimum Mean-Square Error Log-Spectral Amplitude) estimator [11], which estimates the background noise power spectral density (PSD) from the initial unvoiced frames of each clip and applies a frequency-dependent gain function to suppress it. The gain function is bounded by a minimum gain floor to mitigate musical noise and

avoid over-suppression, while a recursive smoothing parameter ($\alpha \approx 0.92$) stabilizes the priori SNR and gain estimates over time. The enhanced magnitude spectrum is recombined with the original noisy phase and reconstructed back into a time-domain waveform via the inverse Short-Time Fourier Transform (iSTFT).

The second denoising strategy uses a U-Net trained from scratch to predict a time-frequency mask that separates speech from noise in the spectrogram. Spectrograms were computed via STFT with an FFT size of 512, hop length of 128, and a 512-sample Hann window. The network follows a three-level encoder-decoder architecture with a base channel width of 16, doubling at each stage, and outputs a sigmoid-activated mask applied to the noisy magnitude spectrogram. Training minimized the L1 loss between the log-mel spectrogram of the denoised and clean waveforms, optimized with AdamW ($\text{lr} = 10^{-3}$, weight decay = 10^{-5} , batch size 8) and cosine-annealing learning rate decay. It was trained for 25 epochs on clean ESC-50 clips from the train fold, mixed with AWGN and traffic noise sample, with the test fold held out entirely. At inference, the predicted mask is applied to the noisy spectrogram, and the masked spectrum is converted back to a waveform via inverse STFT. This follows the spectral masking approach used in U-Net based speech enhancement [4].

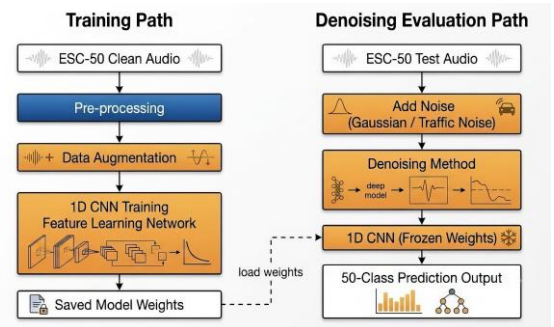


Fig. 1: Overview of the raw waveform-based ESC system.

B. Pipeline 2 : MFCC-Based Subspace Discriminant Ensemble

The proposed MFCC-based classical machine learning pipeline is illustrated in Fig. 2. During training, signal-level data augmentation is applied to improve generalization. The augmentation strategy includes time stretching ($0.85\text{--}1.15\times$), pitch shifting (± 2 semitones), and circular time shifting (± 0.5 s), expanding the training set from 1,600 original samples to approximately 14,400 augmented samples.

Following augmentation, Mel-Frequency Cepstral Coefficients (MFCCs) are extracted with their first order (delta) and second order (delta-delta) derivatives. These features are then summarized through statistical feature aggregation, where each coefficient is characterized using five descriptive measures: mean, standard deviation, maximum, minimum, and median. This results in a fixed-length representation suitable for classical machine learning models.

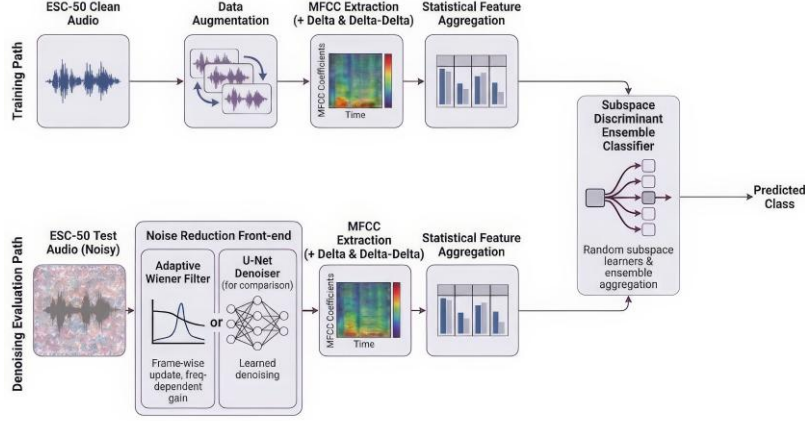


Fig. 2. MFCC-based ESC pipeline showing training-time augmentation and evaluation.

The resulting feature vectors are normalized using standard score normalization to ensure consistent feature scales across all dimensions. The normalized features are subsequently supplied to a Subspace Discriminant Ensemble classifier. The ensemble constructs multiple discriminant learners using randomly selected subsets of the feature space and combines their predictions through ensemble aggregation.

For noise robustness evaluation, an adaptive Wiener filter is employed as a front-end enhancement stage prior to feature extraction. The filter operates in the time domain, applying a per-frame gain computed from local signal variance. The gain function [19] is defined as:

$$G[m] = \max\left(0, 1 - \frac{\hat{\sigma}_n^2}{\sigma_y^2[m]}\right) \quad (2)$$

and the enhanced signal is given by:

$$\hat{x}[n] = G[m] \cdot y[n], \quad n \in \text{frame } m$$

where $\sigma_y^2[m]$ is the signal variance within frame m (512 samples, 256-sample hop, Hanning-windowed), and $\hat{\sigma}_n^2$ is a single noise-variance estimate taken as the minimum frame variance observed across the clip.

Unlike MMSE-LSA, this noise estimate is fixed per clip rather than recursively updated, while the signal-power term is recomputed every frame, allowing the gain itself to track local energy fluctuations without assuming global noise stationarity.

Note that the scalar gain $G[m]$ is mathematically equivalent to the Wiener filter ratio, since

$$1 - \frac{\sigma_n^2}{\sigma_y^2} = \frac{\sigma_y^2 - \sigma_n^2}{\sigma_y^2} \approx \frac{S_x}{S_x + S_n} \quad (3)$$

However, it is computed from time-domain variance rather than per-frequency-bin power spectral density, meaning no FFT is applied. The noise estimate $\hat{\sigma}_n^2$ is computed once per clip specifically as the minimum frame variance across the 5-second signal and reused for every frame's gain calculation.

This directly contrasts with MMSE-LSA, where the

noise estimate is updated recursively via a smoothing factor β . Here, only the local signal variance $\sigma_y^2[m]$ varies per frame, giving the gain function the ability to adapt to local energy fluctuations without a frame-by-frame noise update.

A U-Net spectral-mask denoiser was also trained for this pipeline to provide a deep-learning point of comparison against the statistical methods above. The architecture follows the same approach as Pipeline 1, with an FFT size of 1024, hop length of 256, and base channel width of 32. Training differed from Pipeline 1, however, using a mask-MSE objective rather than the mel-spectrogram L1 loss, optimized with Adam and a plateau-based learning rate schedule for 10 epochs.

C. Pipeline 3: CNN14 with Log-Mel Spectrogram

The proposed CNN14-based pipeline, illustrated in Fig. 3, utilizes log-mel spectrograms as the input representation for ESC. Audio signals were transformed into log-mel spectrograms via framing, Hann windowing, FFT computation, application of a 128-band mel filter bank, and logarithmic compression. The resulting time-frequency image captures perceptually relevant spectral information of the audio signal.

Data augmentation is employed to improve generalization, including circular time shifting ($\pm 10\%$), SpecAugment (time and frequency masking with up to ~ 12 mel bins per mask), Mixup ($\lambda \sim \text{Beta}(0.4, 0.4)$, $p = 0.5$), and CutMix ($p = 0.5$ when Mixup is not applied). Label smoothing ($\epsilon = 0.1$) is applied throughout training.

The augmented log-mel spectrograms were then processed using CNN14 [2], pretrained on the AudioSet dataset for large-scale audio classification. The CNN14 backbone consists of six convolutional blocks, where each block contains two 3×3 Conv2D layers followed by Batch Normalization, ReLU activation, and 2×2 average pooling. The convolutional backbone is followed by a fully connected layer and a lightweight 50-class classification head adapted for the ESC-50 dataset. Dropout ($p=0.2$) is applied after each convolutional block to reduce overfitting.

CNN14 was initialized with AudioSet-pretrained weights [2] and fine-tuned using a three-phase progressive strategy with manually fixed learning rates, following standard transfer-learning practice of assigning smaller rates to earlier, pretrained layers and larger rates to later, task-specific layers [20]. In Phase 1 (15 epochs), only the classification head was trained with the backbone frozen ($\text{lr} = 10^{-3}$). In Phase 2 (40 epochs), the top three convolutional blocks and FC layer were unfrozen using differential rates (5×10^{-6} lower layers, 2×10^{-5} upper layers, 2×10^{-4} classification head). In Phase 3 (20 epochs), the full backbone was unfrozen with a four-group schedule from 10^{-6} to 10^{-4} . AdamW was used throughout, with early stopping (patience = 7) per phase. As shown in Table II, Phase 3 yielded the highest accuracy ($86.95\% \pm 2.5\%$ mean 5-fold CV, peaking at 90.25% on Fold 4), confirming that full-backbone fine-tuning with the smallest rates on the earliest layers best preserved pretrained features while adapting to ESC-50.

Table II: CNN14 accuracy across 3 phases

Fold	Phase 1	Phase 2	Phase 3
1	73.50%	84.25%	85.25%
2	77.25%	88.50%	88.75%
3	73.50%	85.00%	83.75%
4	76.00%	89.50%	90.25%
5	72.75%	85.00%	86.75%
Mean $\pm$ Std	74.6 $\pm$ 1.7%	86.5 $\pm$ 2.0%	86.95$\pm$2.5%

For noise robustness evaluation, the decision-directed MMSE-LSA estimator was applied as a pre-processing step, operating on STFT frames (1024-point FFT, hop length 320, Hann window) at the model's 32 kHz operating rate. Background noise is estimated from the initial 0.3 s segment, and each frame is processed using a frequency-dependent gain function bounded between a 1% floor and 1, which suppresses noise while preserving spectral structure. The priori SNR is recursively estimated using temporal smoothing ($\alpha \approx 0.98$) and the noise PSD is updated at $\beta \approx 0.95$, stabilizing gain adaptation. Alternatively, a deep-learning-based U-Net denoiser was evaluated to provide a learned mapping for noise reduction. The network architecture, loss function, and optimization setup matched Pipeline 1.

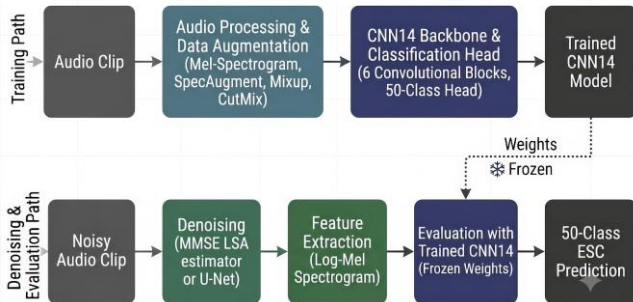


Fig.3. Structural pipeline flow for noisy audio processing, feature representation, and robust CNN14-based sound classification.

V. EXPERIMENTAL RESULTS

A. Clean Audio Baseline

Table III summarizes the classification performance of the three baseline pipelines evaluated on clean audio. For the clean baseline, the results were obtained using 5-fold cross-validation according to the ESC-50 dataset's predefined fold structure, with the reported mean $\pm$ standard deviation computed across the five folds. The pretrained CNN14 model achieves the highest overall accuracy ($86.95\pm2.5\%$), significantly outperforming the MFCC-based ensemble ($54.00\%\pm0.74$) and the 1D CNN model ($48.34\% \pm 4.8$). This observed performance hierarchy directly reflects the representational capacity inherent to each architectural paradigm. Operating on log-mel spectrograms, the pretrained CNN14 leverages deep, hierarchical feature representations optimized via large-scale pretraining [2]. Conversely, while the MFCC ensemble benefits from robust subspace aggregation, its expressiveness is fundamentally constrained by the fixed nature of handcrafted cepstral descriptors [1]. The 1D CNN, processing raw waveforms, exhibits the lowest baseline performance, highlighting the difficulty of learning robust feature detectors from scratch given the limited sample size of the ESC-50 dataset.

TABLE III: Clean audio classification performance on ESC-50. Results are reported as mean $\pm$ standard

Model	Acc.	Prec.	Rec.	F1
1D CNN	48.34% $\pm$ 4.8	53.9%	47.5%	46.7%
MFCC Ensemble	54.00% $\pm$ 0.74	57.7%	55.8%	54.4%
CNN14 (PANNs)	86.95$\pm$2.5%	89.4%	87.5%	86.8%

B. Noise Robustness Experiment

1. Raw waveform

Under synthetic Gaussian noise, the statistical MMSE-LSA denoiser provides stable classification improvements across high SNR levels, yielding accuracy gains between +4.2% and +10.3%. Under real-world traffic noise, MMSE-LSA also improved performance at most SNR levels, achieving a maximum recovery of +15.45% at 5 dB SNR. However, performance slightly degraded -10 dB SNR.

In contrast, as shown in Table IV, the learned U-Net denoiser yields substantially larger recovery metrics across all testing conditions. The U-Net achieves its most pronounced advantages at lower SNRs, recovering +37.30% at -5 dB SNR and +36.05% at 0dB SNR under real-world traffic conditions.

These traffic-noise gains were statistically significant at every tested level ($p < 0.0001$, paired t-test, $n = 5$ runs).

TABLE IV: 1D CNN classification accuracy and recovery performance under Gaussian and traffic noise conditions for the U-Net denoiser

SNR	Noisy	U-Net	Recovery
Gaussian Noise			
10 dB	23.2% $\pm$ 0.3	45.6% $\pm$ 0.5	+22.4%
5 dB	16.4% $\pm$ 0.3	39.4% $\pm$ 0.4	+23.0%
0 dB	10.5% $\pm$ 0.0	27.0% $\pm$ 0.32	+16.5%
-5 dB	6.5% $\pm$ 0.4%	13.1% $\pm$ 0.4	+6.6%
-10 dB	2.4% $\pm$ 0.1%	4.3% $\pm$ 0.0	+1.9%
Traffic Noise			
10 dB	29.8% $\pm$ 0.33	48.1% $\pm$ 0.58	+18.30%
5 dB	21% $\pm$ 0.47	47.8% $\pm$ 0.37	+26.80%
0 dB	10.20% $\pm$ 0.37	46.25% $\pm$ 1.02	+36.05%
-5 dB	2.75% $\pm$ 0.0	40.05% $\pm$ 0.82	+37.30%
-10 dB	3.35% $\pm$ 0.22	21.7% $\pm$ 0.78	+18.35%

2. MFCC

The Adaptive Wiener filter provides consistent positive recovery across Gaussian noise at all tested SNR levels. At SNR = 5 dB white DSP yields 21.33% vs. 10.50% without (+10.83%, $p = 0.0011$). Critically, applying MMSE-LSA to the MFCC pipeline instead of the Adaptive Wiener filter degrades performance across all tested conditions the strongest empirical evidence for the denoiser representation interaction.

However, as shown in *Table V*, The U-Net denoiser achieved significant gains under Gaussian noise, recovering +12.25 to +26.60 percentage points ($p \leq 0.002$ at all SNR levels). Under traffic noise, gains of +9.75 to +17.75 percentage points were also significant ($p = 0.003$ – 0.038), except at 5 dB, which was borderline ($p = 0.050$). A Wilcoxon test confirmed consistent directionality but, with U-Net winning all 5 runs at every level, hit its minimum attainable value ($p = 0.0625$) uniformly and offered no further resolution.

TABLE V: MFCC ensemble classification accuracy and recovery performance under Gaussian and traffic noise conditions for the U-Net denoiser

SNR	Noisy	U-Net	Recovery
Gaussian Noise			
10 dB	22.55% $\pm$ 0.60	41.35% $\pm$ 4.81	+18.80%
5 dB	9.00% $\pm$ 0.40	35.6% $\pm$ 4.43	+26.60%
0 dB	2.95% $\pm$ 0.21	27.55% $\pm$ 3.50	+24.60%
-5 dB	2.05% $\pm$ 0.11	20.20% $\pm$ 3.56	+18.15%
-10 dB	2.00% $\pm$ 0.00	14.25% $\pm$ 4.02	+12.25%
Traffic Noise			
10 dB	26.85% $\pm$ 1.10	33.10% $\pm$ 5.28	+6.25%
5 dB	19.00% $\pm$ 0.65	36.75% $\pm$ 4.86	+17.75%
0 dB	11.55% $\pm$ 0.91	28.05% $\pm$ 3.58	+16.50%
-5 dB	5.50% $\pm$ 0.80	18.25% $\pm$ 2.75	+12.75%
-10 dB	2.25% $\pm$ 0.33	12% $\pm$ 2.08	+9.75%

3. CNN14

Under Gaussian noise, the MMSE-LSA denoiser provided consistent improvements across the evaluated SNR range, achieving a maximum recovery of +24.25% at -5 dB SNR. However, under real-world traffic noise, MMSE-LSA significantly degraded performance, with the largest reduction observed at 0 dB SNR, where accuracy dropped from 61.75% to 35.90% (-25.85% , $p < 0.05$).

The U-Net denoiser avoids these vulnerabilities, demonstrating uniform robustness across both noise types. As outlined in *Table VI*, the U-Net delivers high recovery gains under traffic noise at low SNRs, achieving a +24.30% accuracy recovery at -5 dB SNR and +21.50% at -10 dB SNR.

TABLE VI: CNN14 classification accuracy and recovery performance under Gaussian and traffic noise conditions for the U-Net denoiser

SNR	Noisy	U-Net	Recovery
Gaussian Noise			
10 dB	69 $\pm$ 0.47	75.65 $\pm$ 1.64	+6.65%
5 dB	52.30 $\pm$ 0.65	61.45 $\pm$ 0.57	+9.15%
0 dB	28.10 $\pm$ 0.68	43.10 $\pm$ 1.13	+15.00%
-5 dB	8.90 $\pm$ 0.45	23.30 $\pm$ 1.24	+14.40%
-10 dB	2.15 $\pm$ 0.22	11.75 $\pm$ 0.35	+9.60%
Traffic Noise			
10 dB	79.25 $\pm$ 0.59	87.20 $\pm$ 0.72	+7.95%
5 dB	69.15 $\pm$ 0.49	83.15 $\pm$ 0.80	+14.00%
0 dB	55.55 $\pm$ 0.78	76.30 $\pm$ 1.22	+20.75%
-5 dB	36.65 $\pm$ 0.72	60.95 $\pm$ 0.72	+24.30%
-10 dB	18.65 $\pm$ 0.52	40.15 $\pm$ 1.57	+21.50%

As illustrated in *Fig. 4*, the changes in the spectrograms match our numerical results. Even though traffic noise spans the entire spectrum, the U-Net successfully removes the background noise while preserving the sharp lines, transient edges, and key shapes of the primary sound events

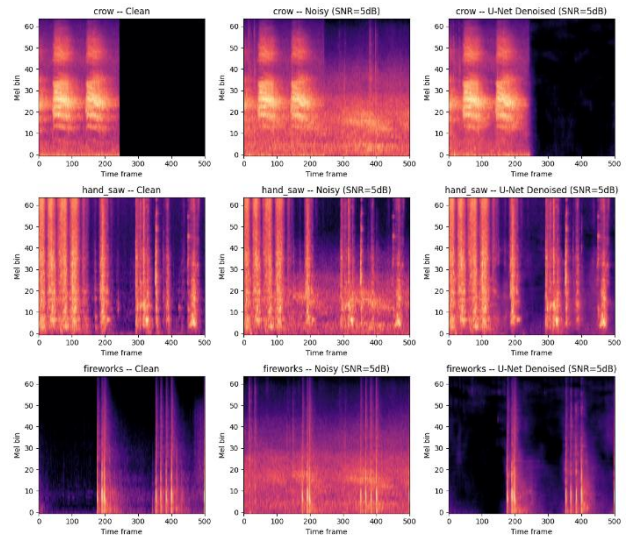


Fig 4: Mel-spectrograms of clean, traffic noise (SNR = 5 dB), and U-Net denoised audio for crow, hand saw, and fireworks, showing that U-Net suppresses noise while retaining key spectral structure.

VI. DISCUSSION

The experimental results validate our core thesis: a denoiser cannot be evaluated as an isolated pre-processing step. Its success depends entirely on how the downstream model structures and processes the audio features. Across all three pipelines, the exact same denoising methods produced completely different classification outcomes depending on the feature representation used and the type of noise. A systematic analysis of these interactions reveals distinct architectural behaviors:

Feature Preservation vs. Perceptual Quality: The MFCC-based pipeline (Table V) is the most sensitive to processing artifacts. While the adaptive Wiener filter provides small, stable gains via steady frame-by-frame updates, the rigid gain functions of MMSE-LSA consistently degrade performance. Because MFCC extraction relies heavily on smooth spectral envelopes and statistical derivatives, it is highly vulnerable to the "musical noise" and artificial smoothing introduced by traditional statistical estimators. This proves a critical point: speech enhancement techniques designed to make audio sound cleaner to human ears do not necessarily preserve the structural features that a classifier needs to recognize a sound.

The Stationarity Assumption Bottleneck: A traditional denoiser only works well if its mathematical assumptions match the actual environment. This is clearly shown by MMSE-LSA within the CNN14 pipeline (Table VI). Under stationary AWGN, where the background noise does not change over time, the algorithm successfully protects the macro-level structures in the log-mel spectrogram. However, under non-stationary traffic noise, the background environment changes constantly. This violates the algorithm's local stationarity assumptions, causing the tracker to introduce severe spectral distortions. These artifacts destroy the visual patterns in the log-mel spectrogram, leading to the sharp performance drop observed at 0 dB SNR.

The Robustness of Data-Driven Mapping: In stark contrast to classical DSP estimators, the learned U-Net denoiser generalizes well across all three pipelines and both noise types. As shown in the spectrograms of Fig. 4, the U-Net effectively removes background noise without erasing the sharp edges and brief, transient sound events (like the crow or fireworks) that the classifiers rely on. This clear visual preservation explains the high numerical recoveries seen in *Tables IV, V, and VI*. Ultimately, features like MFCCs and log-mel spectrograms benefit far more from the continuous time-frequency masks learned by neural networks than from traditional statistical spectral subtraction.

Consequently, these results demonstrate that building a noise-robust ESC system requires a co-designed approach. Rather than treating the denoiser, the noise

type, and the feature representation as separate stages, they must be optimized together as parts of a unified system.

VII. CONCLUSION

This paper presented a systematic investigation of denoiser–representation interactions in noise-robust environmental sound classification using the ESC-50 dataset. Three classification pipelines based on raw waveforms, MFCC features, and log-mel spectrograms were evaluated under Gaussian and real-world traffic noise across a wide range of SNR conditions. The results demonstrated that denoising effectiveness depends not only on the enhancement method itself, but also on the downstream feature representation and the characteristics of the noise environment.

Classical enhancement methods, including MMSE-LSA and adaptive Wiener filtering, improved performance when their underlying assumptions aligned with the representation and noise conditions, but could also degrade classification accuracy when this alignment was violated. In contrast, the learned U-Net denoiser consistently achieved the largest recovery gains across all three pipelines and exhibited superior robustness under non-stationary traffic noise. These findings indicate that denoising and feature extraction should not be treated as independent design stages, but rather as jointly interacting components of an environmental sound classification system.

Future work will investigate end-to-end optimization of denoising and classification models, evaluate additional learned enhancement architectures, and extend the analysis to larger real-world acoustic datasets and more diverse environmental noise conditions

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